

CLIMATE CHANGE TWITTER SENTIMENT ANALYSIS

End-to-End Data Analysis SQL & BI Project Report

By

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Tools Used: PostgreSQL 18 · Power BI Desktop · Power BI Service

Dataset: The Climate Change Twitter Dataset (Mendeley Data, 2022)

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1. INTRODUCTION AND DATASET OVERVIEW

This report presents an end-to-end data analysis of the Climate Change Twitter Dataset, comprising 15,789,411 tweets spanning 14 years from 2006 to 2019. The analysis was commissioned to help a climate research organisation understand two core questions ahead of their next policy briefing:

1. How have public sentiment and climate change stance shifted over 14 years?
2. Which topics and regions are driving the most divisive or aggressive discourse?

The full pipeline was executed using PostgreSQL 18 for data ingestion, cleaning, transformation, and analytical querying, and Power BI for interactive visualisation and dashboard development.

1.1. Dataset overview

Feature	Details
Dataset	The Climate Change Twitter Dataset.csv
Total Columns	10
Total Rows	15,789,411

1.2 Data Dictionary

Column	Description
created_at	Timestamp of tweet posting
id	Unique tweet identifier
lat	Latitude coordinate of tweet location
lng	Longitude coordinate of tweet location

topic	Climate topic category assigned to tweet
sentiment	Polarity score of tweet content
stance	Author's inferred stance on climate change
gender	Algorithmically inferred author gender
temperature_avg	Local temperature deviation in °C at tweet location
aggressiveness	Model-assigned aggressiveness label

1.3 Derived Columns Added During Cleaning:

Column	Description
tweet_date	Date portion of tweet_timestamp
tweet_year	Year extracted from timestamp
tweet_month	Month extracted from timestamp
tweet_quarter	Quarter extracted from timestamp
tweet_day_of_week	Day of week (0=Sunday to 6=Saturday)
has_coordinates	TRUE when both lat and lng are populated
sentiment_category	Positive (>0.05) / Negative (<-0.05) / Neutral
is_aggressive	TRUE when aggressiveness = Aggressive
continent	Derived from lat/lng bounding boxes

2. METHODOLOGY

2.1 Approach Overview

The project followed a structured five-phase pipeline:

Phase 1 (Raw Ingestion): The CSV dataset was imported into PostgreSQL 18 as a raw **staging table** with all columns stored as TEXT (except id which was auto-detected as BIGINT). This approach guaranteed a successful import regardless of edge cases such as scientific notation, unexpected blanks, or formatting inconsistencies in the source data. The TEXT-first approach is the professional standard for large datasets where column contents are unknown upfront.

Phase 2 (Data Quality Audit): Every column was systematically audited for nulls, blanks, non-numeric values, out-of-range values, unexpected categories, and duplicate records before any cleaning logic was written. All findings were documented and informed the cleaning decisions in Phase 3.

Column	Raw Type	Cleaned Type	Description	Values / Range	Cleaning Applied
created_at	TEXT	TIMESTAMPTZ	Timestamp of tweet posting	2006-06-06 to 2019-12-31	Cast to TIMESTAMPTZ; year, month, quarter, day_of_week extracted
id	BIGINT	BIGINT	Unique tweet identifier	6,132 to 1,178,911,891,285,917,696	Used as primary key; 0 nulls confirmed

lat	TEXT	NUMERIC(10, 7)	Latitude coordinate of tweet location	-90 to +90; 66.4% blank	Blank set to NULL; cast to NUMERIC
lng	TEXT	NUMERIC(10, 7)	Longitude coordinate of tweet location	-180 to +180; 66.4% blank	Blank set to NULL; 2 scientific notation values cast correctly
topic	TEXT	VARCHAR(100)	Climate topic category assigned to tweet	10 distinct values	INITCAP and TRIM applied for consistency
sentiment	TEXT	NUMERIC(10, 7)	Polarity score of tweet content	-0.994 to +0.992	Cast to NUMERIC; categorised into Positive/Neutral/Negative
stance	TEXT	VARCHAR(20)	Author's inferred stance on climate change	believer / denier / neutral	Title-cased to Believer / Denier / Neutral
gender	TEXT	VARCHAR(20)	Algorithmically inferred author gender	male / female / undefined	undefined and NULL mapped to Unknown

temperature_avg	TEXT	NUMERIC(10, 7)	Local temperature deviation in °C at tweet location	-23.3 to +21.0; 66.4% blank	Blank set to NULL; 37 scientific notation values cast correctly
aggressiveness	TEXT	VARCHAR(20)	Model-assigned aggressiveness label	aggressive / not aggressive	Title-cased; binary is_aggressive flag added

Phase 3 (Data Cleaning and Transformation): A clean typed table (clean_tweets) was created with proper data types, derived columns, and standardised categorical values. The INSERT was optimised using a subquery pattern to cast each numeric column once and reuse the result, avoiding repeated casting on 15.7 million rows.

Phase 4 (Analytical Views): Twelve pre-aggregated views were created in PostgreSQL to serve as the data source for Power BI. Each view was designed to answer a specific analytical question and includes detailed SQL comments explaining the business logic.

Phase 5 (Visualisation and Dashboarding): Power BI Desktop was connected directly to the PostgreSQL views using the Npgsql driver on port 5433. An interactive four-page dashboard was built and published to Power BI Service.

2.2 Tools and Technologies

Tool	Version	Purpose
PostgreSQL	18	Data storage, cleaning, transformation, analytics
pgAdmin	4	PostgreSQL GUI and query execution
Power BI Desktop	Latest	Dashboard development
Power BI Service	Cloud	Dashboard publishing and sharing

2.3 Sentiment Categorisation Thresholds

Sentiment scores were categorised using the following thresholds:

- **Positive:** score > +0.05
- **Neutral:** score between -0.05 and +0.05
- **Negative:** score < -0.05

These thresholds create a meaningful neutral band around zero, avoiding misclassification of marginally positive or negative scores.

2.4 Divisiveness Score Formula

A divisiveness score was computed for each topic and region to measure how evenly split discourse is between believers and deniers:

$$\text{divisiveness_score} = 1 - |\text{believer_ratio} - 0.5| \times 2$$

Where $\text{believer_ratio} = \text{believers} / (\text{believers} + \text{deniers})$, excluding neutral tweets.

A score of 1.0 indicates a perfectly even 50/50 split (maximally divisive). A score of 0.0 indicates all tweets are from one side (no division).

2.5 Continent Classification

The continent was derived from latitude and longitude coordinates using approximate bounding boxes. This is a simplified proxy for geolocation and may misclassify border regions such as Eastern Europe or the Middle East. It is used for high-level regional trend analysis only.

3. DATA QUALITY FINDINGS

3.1 Pre-Cleaning Audit Results

Column	Total Rows	Nulls/Blanks	Non-Numeric	Out of Range	Duplicates
created_at	15,789,411	0	0	0	—
id	15,789,411	0	0	—	0
lat	15,789,411	10,481,873 (66.4%)	0	0	—
lng	15,789,411	10,481,873 (66.4%)	2 (scientific notation)	0	—
sentiment	15,789,411	0	0	0	—
temperature_avg	15,789,411	10,481,873 (66.4%)	37 (scientific notation)	0	—
stance	15,789,411	0	—	—	—
gender	15,789,411	0	—	—	—
aggressiveness	15,789,411	0	—	—	—
topic	15,789,411	0	—	—	—

3.2 Notable Findings

Scientific Notation Values: Two longitude values and 37 temperature values were stored in scientific notation (e.g., $-5.22e-05$). These are valid near-zero numbers and were correctly handled by PostgreSQL's native NUMERIC casting.

Missing Geolocation: 66.4% of tweets have no coordinates. This is expected for Twitter data where location sharing is optional. Regional analysis covers only the 33.6% of geotagged tweets (5,307,538 rows).

Gender Undefined: 586,875 tweets (3.7%) have gender labelled as undefined. These were mapped to Unknown during cleaning.

Zero Duplicates: All 15,789,411 tweet IDs are unique — no deduplication was required.

Date Consistency: All timestamps follow a consistent format (YYYY-MM-DD HH:MM:SS+00:00) with no malformed dates.

3.3 Post-Cleaning Validation

Metric	Value
Rows in raw_tweets	15,789,411
Rows in clean_tweets	15,789,411
Rows lost during cleaning	0
Null sentiment scores	0
Valid coordinates	5,307,538 (33.6%)
Unknown gender	586,875 (3.7%)
Unknown stance	0

Unknown continent	10,481,873 (66.4%)
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4. KEY FINDINGS

Key findings at a glance:

- Climate change believers grew from 16.7% in 2006 to 79.4% in 2018, a dramatic shift in public opinion over 14 years.
- The Paris Agreement year (2015) marked a clear turning point , average sentiment shifted from consistently negative (-0.025) to consistently positive (+0.005).
- Donald Trump versus Science is the most divisive topic (divisiveness score: 0.57) and Politics is the most aggressive (43.4% of tweets flagged aggressive).
- North America drives 65% of all geotagged discourse and is the most aggressive region at 31.8%.
- Deniers are 56% more aggressive than Believers (42.7% vs 27.3%).
- Aggressiveness declined steadily from 35% in 2007 to 24% in 2019, an encouraging long-term trend

4.1 Sentiment and Stance Shifted Dramatically Over 14 Years

The most significant finding of this analysis is the dramatic shift in public stance on climate change between 2006 and 2019. Climate change believers grew from just 16.7% of discourse in 2006 to a peak of 79.4% in 2018. Simultaneously, deniers declined from a peak of 14.3% in 2010 to just 6.2% in 2018.

Average sentiment followed a similar trajectory. From 2006 to 2014, average annual sentiment was consistently negative, ranging from -0.049 to -0.009. From 2015 onwards, sentiment turned predominantly positive, peaking at +0.046 in 2016.

The Paris Agreement year (2015) represents the clearest turning point in the data. Pre-Paris average sentiment (2006-2014) was -0.025 compared to post-Paris (2016-

2019) at +0.005. Believer percentage jumped from 45.0% in 2014 to 68.3% in 2015 — the single largest year-on-year increase in the dataset.

Tweet volume grew exponentially, from just 12 tweets in 2006 to a peak of 6,259,190 in 2018, reflecting the explosive growth of Twitter as a platform and increasing public engagement with climate issues.

4.2 Most Divisive and Aggressive Topics

Donald Trump versus Science is the most divisive topic with a divisiveness score of 0.57, indicating the most evenly contested discourse between believers and deniers. It is also the second most aggressive topic with 40.3% of tweets flagged as aggressive and has the second most negative average sentiment at -0.160.

Politics is the most aggressive topic overall at 43.4% aggressiveness despite having a relatively low divisiveness score of 0.15, suggesting that political climate discourse is hostile but not evenly contested — it is dominated by believers who express their views aggressively.

Ideological Positions on Global Warming has the most negative average sentiment at -0.198, indicating it generates the most pessimistic or hostile language of all topics.

Undefined/One Word Hashtags has the most positive average sentiment at +0.241, suggesting that casual climate-adjacent content is significantly more positive than substantive climate discourse.

Topic dominance shifted significantly over time. Weather Extremes dominated discourse from 2007 to 2014, while Global Stance became the top topic from 2015 onwards, reflecting a shift from event-driven to opinion-driven climate discourse. Politics and Donald Trump versus Science emerged strongly from 2017, driven by US political developments.

4.3 Regional Patterns

North America dominates geotagged climate discourse, accounting for 65.1% of all tweets with coordinates. It is also the most aggressive region at 31.8% and the most divisive at a score of 0.20.

The North America + Politics combination is the single most aggressive region–topic combination in the dataset at 45.8%, followed by North America + Donald Trump versus Science at 43.4%. This points to US political polarisation as a primary driver of aggressive climate discourse globally.

Europe is the second most active region at 19.5% of geotagged tweets and has the highest average temperature deviation at +2.13°C, yet its aggressiveness rate (22.9%) is lower than both North America and Oceania.

Africa and Asia show the most positive average sentiment at +0.141 and +0.144 respectively despite being regions particularly vulnerable to climate impacts. Both also have the lowest aggressiveness rates, suggesting more constructive engagement with climate topics in these regions.

4.4 Correlation Analysis

Pearson correlation coefficients were calculated between the key numeric variables:

Variables	Correlation	Interpretation
Sentiment vs Aggressiveness	-0.1603	Moderate negative – more negative sentiment leads to more aggression
Temperature vs Sentiment	+0.0174	Near zero – temperature deviation barely affects sentiment
Temperature vs Aggressiveness	-0.0094	Essentially no relationship

The near-zero correlation between temperature deviation and both sentiment and aggressiveness is a significant finding. It suggests that climate discourse is driven by political and media cycles rather than actual climate events. Higher local temperatures do not make people angrier or more negative about climate change on Twitter.

4.5 Demographic Patterns

By Stance: Deniers are significantly more aggressive than Believers. 42.7% of Denier tweets are flagged as aggressive compared to 27.3% of Believer tweets — a 56% higher rate. Deniers also have the most negative average sentiment at -0.232 compared to Believers at $+0.017$.

By Gender: Males are slightly more aggressive than Females at 29.5% vs 27.4%. The difference is modest but consistent across years.

By Season: Summer is the most aggressive season at 31.1% while Spring is the least aggressive at 27.7%. Winter shows the most negative average sentiment at -0.003 compared to Autumn at $+0.009$.

4.6 Long-Term Aggressiveness Decline

One of the most encouraging findings in the data is the consistent long-term decline in aggressiveness. The overall aggressiveness rate fell from 35.1% in 2007 to 24.2% in 2019 — a reduction of approximately 11 percentage points over 12 years. This decline accelerated post-2017, suggesting that as climate change became more mainstream and accepted, discourse became less hostile.

5. RECOMMENDATIONS

Based on the analytical findings, the following six data-driven recommendations are made for the climate research organisation's policy briefing:

Target Counter-Messaging at North America: North America accounts for 65% of geotagged climate discourse and has the highest aggressiveness rate at 31.8%. The North America + Politics and North America + Donald Trump versus Science combinations are the two most aggressive region-topic hotspots in the entire dataset. Communication and counter-messaging campaigns should prioritise this region, particularly around political and Trump-related climate topics.

Leverage the Paris Agreement Effect: The 2015 Paris Agreement produced the single largest year-on-year shift in both sentiment (+0.034) and believer percentage (+23.3 percentage points) in the dataset. This demonstrates that high-profile international policy events directly and measurably shift public opinion. Future policy interventions and international agreements should be accompanied by coordinated public communication campaigns to amplify this effect.

Focus Moderation on Denier Communities: Deniers generate aggressiveness at a 56% higher rate than Believers and consistently produce the most negative sentiment. They represent the primary source of toxic climate discourse on Twitter. Platform moderation efforts and de-escalation strategies should be specifically targeted at denier-dominated conversations, particularly in the Donald Trump versus Science and Ideological Positions on Global Warming topic categories.

Recommendation 4 — Monitor Donald Trump versus Science as a High-Risk Topic

This topic is simultaneously the most divisive (0.57) and second most aggressive (40.3%) in the dataset. It emerged strongly from 2017 and is concentrated in North America. The organisation should establish dedicated monitoring for this topic category to track escalation and identify key influencers driving divisive discourse.

Reframe Communication Around Opinion Not Events: The near-zero correlation between temperature deviation and both sentiment and aggressiveness confirms that people do not respond to actual climate data on Twitter. Discourse is driven by political events and media cycles. The organisation should invest in narrative and

opinion-led communication strategies rather than data-driven weather-event reporting. The shift from Weather Extremes to Global Stance as the dominant topic from 2015 onwards supports this recommendation.

Highlight the Positive Long-Term Trend: The consistent decline in aggressiveness from 35% in 2007 to 24% in 2019, combined with the growth of believers from 40% to 79%, represents a genuinely positive story for climate communication. Policy briefings should include this trend prominently to demonstrate that long-term communication efforts are working and to build support for continued investment in climate communication programmes.

6. DASHBOARD GUIDE

The interactive Power BI dashboard is organised across four pages. The public link is available at: [POWER BI PUBLIC LINK](#)

Page 1 — Executive Overview

Provides a high-level summary of the entire dataset. Contains six KPI cards showing total tweets, believer percentage, denier percentage, aggressiveness rate, average sentiment, and geotagged coverage. Charts show sentiment trend with aggressiveness overlay, tweet volume over time with sentiment line, believer vs denier percentage over time, and stance distribution by year.

Slicers: Year range (2006–2019)

Page 2 — Topic Analysis

Deep dive into the 10 topic categories. Charts show tweet volume by topic, aggressiveness rate by topic, divisiveness score by topic, a scatter plot of sentiment vs divisiveness sized by tweet volume, and topic volume trends over time.

Slicers: Topic dropdown

Page 3 — Regional Analysis

Geographic breakdown of climate discourse. Charts show tweet distribution by continent, aggressiveness by continent, sentiment by continent, stance by region over time, and temperature deviation vs sentiment trend.

Slicers: Continent

Page 4 — Demographics and Aggressiveness

Demographic breakdown and aggressiveness deep dive. Charts show aggressiveness by stance, aggressiveness by gender, gender sentiment over time, seasonal aggressiveness patterns, and the Pre/Post Paris Agreement comparison.

Slicers: Year range

Connected Views

The dashboard connects directly to the following PostgreSQL views:

- vw_kpi_summary, vw_yearly_trends, vw_monthly_trends
- vw_topic_analysis, vw_topic_yearly_trends
- vw_regional_analysis, vw_stance_by_region
- vw_gender_analysis, vw_temp_sentiment
- vw_aggressiveness_analysis, vw_executive_insights

7. LIMITATIONS

The following limitations should be considered when interpreting and acting on the findings of this analysis:

Geographic Coverage: 66.4% of tweets have no geographic coordinates, meaning regional analysis covers only 33.6% of the dataset. The geotagged subset may not be representative of the full geographic distribution of climate discourse.

Inferred Gender: Gender labels are algorithmically inferred rather than self-reported, introducing classification error. The 3.7% undefined rate and the binary Male/Female categorisation do not capture the full spectrum of gender identity.

Platform Bias: Twitter's user base skews younger, more educated, more urban, and more English-speaking than the general global population. Findings reflect Twitter discourse and should not be generalised to represent broader public opinion without caution.

Model-Generated Labels: The stance, aggressiveness, and sentiment labels are generated by machine learning models and may contain classification errors. The accuracy of these labels was not independently validated as part of this analysis.

Temporal Coverage: The dataset covers 2006 to 2019 and does not include the COVID-19 period (2020 onwards) which significantly changed social media discourse patterns globally.

Continent Bounding Boxes: Continent classification uses approximate geographic bounding boxes and may misclassify tweets from border regions such as Eastern Europe, the Middle East, or island nations in the Pacific.

8. REFERENCES

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This report was prepared as part of an end-to-end data analysis project. All findings are derived from the Climate Change Twitter Dataset and are subject to the limitations described in Section 7.